

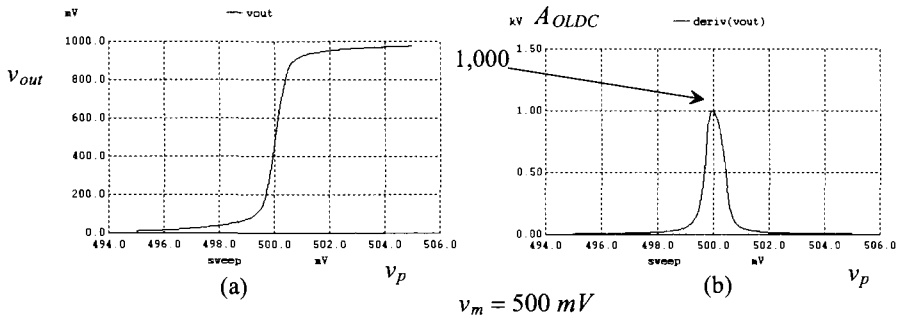


Figure 24.3 (a) DC transfer curves for the op-amp in Fig. 24.2 and (b) its gain (the derivation of (a)).

Power Dissipation

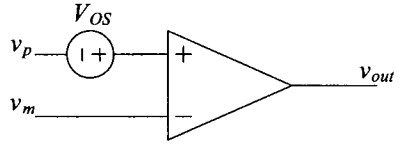
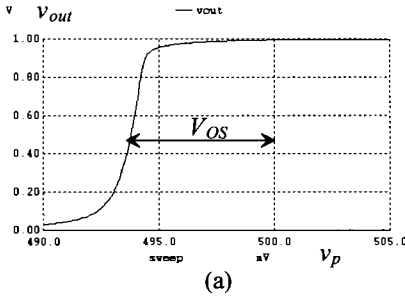
To determine the power dissipated by the op-amp, we sum the currents supplied by the constant current sources and multiply the result by V_{DD} . For the op-amp in Fig. 24.2, the current through M6 is 20 μA and the current through M8 is 10 μA . The total power dissipation is then 30 μW ($V_{DD} = 1\text{ V}$).

Output Swing and Current Sourcing/Sinking Capability

The maximum output swing (for the op-amp in Fig. 24.2) is limited by M7 going into the triode region. If we must keep at least 100 mV across M7, then the maximum output voltage is 900 mV. When M8 goes into triode or roughly 100 mV (see Fig. 20.48), the minimum output voltage is set. As seen in Fig. 24.3a, the high-gain (or large slope) region falls between v_{out} of 100 and 900 mV. Note that the maximum amount of current that this op-amp can sink is limited by the constant current sink M8 or 10 μA . The op-amp can source considerably more than 10 μA by pulling the gate of M7 downwards. Because this topology can source a considerable amount of current, it is useful as a voltage regulator (because the op-amp is always only sourcing current in a voltage regulator application).

Offsets

We've talked in great length (earlier in the book) about random offsets and how to design and layout circuits to minimize their effects. Another type of offset (that is not random) is termed a *systematic offset*. When we sized the MOSFETs in Fig. 24.2, for example, we made sure that M7 was sized to source 10 μA and M8 was sized to sink 10 μA . What would happen if we sized M7 to source 100 μA of current instead (changed its size from 100/2 to 1000/2)? Because M8 is a constant bias of 10 μA , M7 would move into the triode region until it was sourcing the 10 μA of current that M8 wants to sink. The output voltage would be very close to V_{DD} with M7 in the triode region. Effectively we would get a shift or offset in the transfer curves seen in Fig. 24.3a (see Fig. 24.4a). To model this output voltage shift, we can refer it back to the op-amp input as an input-referred offset voltage, Fig. 24.4b. Note that unlike a random offset, which may be positive or negative in value, a systematic offset will always be a known polarity. Finally, note that while we chose to model the offset in series with the noninverting input terminal, in Fig. 24.4, we could just as easily have placed it in series with the inverting op-amp terminal (with a change in polarity).



(b) How the offset is modeled

$$v_m = 500 \text{ mV}$$

Figure 24.4 Showing how increasing M7's width to 1000 in Fig. 24.2 causes an input-referred (systematic) offset voltage.

Compensating the Op-Amp

An important step in op-amp design is the design of the compensation network. The op-amp takes the difference between the inverting and noninverting input terminal voltages and, ideally, multiplies the result by a very large number (the gain). A block diagram representation of an op-amp is seen in Fig. 24.5. The open-loop gain as a function of frequency is labeled $A_{OL}(f)$.

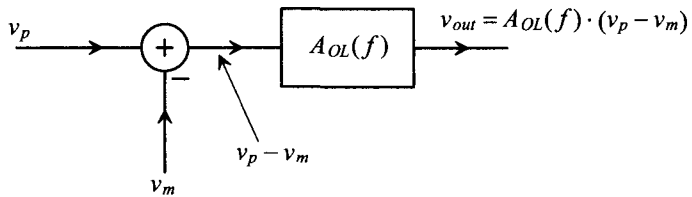


Figure 24.5 Block level diagram of an op-amp.

In all practical situations, the op-amp is used with feedback, Fig. 24.6. While our op-amp in Fig. 24.2 cannot drive a resistive load (unless it is megaohms, as a resistor on the op-amp output will kill the gain of the second stage), we'll still use this figure (Fig. 24.6) to illustrate the concept of feedback and compensation. We can write

$$v_{out} = A_{OL}(f) \cdot (v_{in} - v_f) \quad (24.2)$$

and

$$v_f = v_{out} \cdot \frac{R_2}{R_1 + R_2} \quad (24.3)$$

The amount of the output that is fed back is often called the feedback factor β or

$$\beta = \frac{R_2}{R_1 + R_2} \quad (24.4)$$

Substituting Eqs. (24.3) and (24.4) into Eq. (24.2) and solving for the closed loop gain gives

$$A_{CL}(f) = \frac{v_{out}}{v_{in}} = \frac{A_{OL}(f)}{1 + \beta \cdot A_{OL}(f)} \quad (24.5)$$

If we take $A_{OL}(f) \rightarrow \infty$, then the closed loop gain of this non-inverting topology is

$$A_{CL}(f) \rightarrow \frac{1}{\beta} = 1 + \frac{R_1}{R_2} \quad (24.6)$$

We have several important points that we need to discuss. To begin, notice that in Eq. (24.5) if

$$\beta \cdot A_{OL}(f) = -1 \quad (24.7)$$

or more precisely

$$|\beta \cdot A_{OL}(f)| = 1 \text{ and } \angle \beta \cdot A_{OL}(f) = \pm 180^\circ \quad (24.8)$$

the closed loop gain blows up (the feedback amplifier becomes unstable). The worst case situation (the largest value of β) occurs when all of the output is fed back to the op-amp input (assuming no transformer, amplifier, etc. in the feedback path). The voltage follower, Fig. 24.7, is an example of this situation. *To determine the stability of an op-amp, we'll look at the open loop gain when the feedback factor is one, that is,*

$$|A_{OL}(f)| = 1 \text{ and } \angle A_{OL}(f) = 180^\circ \quad (24.9)$$

Notice that the larger the closed loop gain, the smaller the value of β (the less output signal we feed back) and the more likely the op-amp circuit, with feedback, will be stable. *This is important.* While feedback helps to desensitize an amplifier's gain to variations in an op-amp's A_{OL} , the drawback is stability. In a high-performance op-amp that will never operate in a unity-follower configuration, we can get an enhancement in speed by reducing the amount of compensation (the amount of the reduction guided by Eq. (24.8) with the actual value of β used). While this discussion has focused on the non-inverting topology (voltage or series-shunt amplifiers, $\beta = v_f/v_{out}$) a similar discussion can be given for the inverting amplifier (shunt-shunt where $\beta = i_f/v_{out}$) topology, see Sec. 30.2.

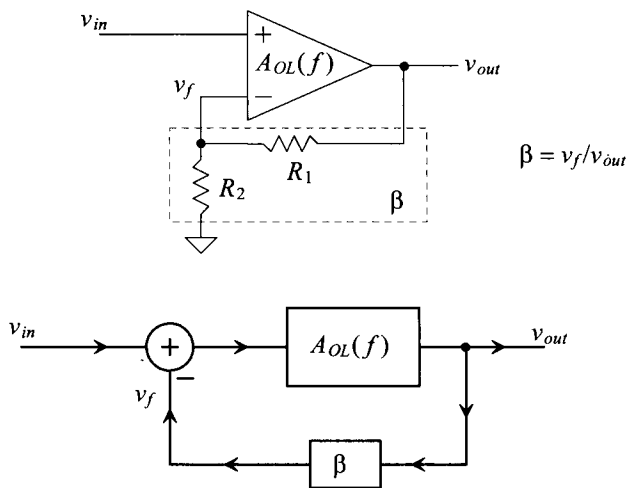


Figure 24.6 An example of feedback in an op-amp, see Sec. 30.2 for additional discussion.

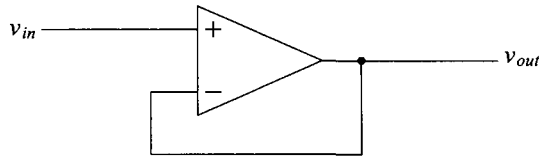


Figure 24.7 Voltage follower configuration, an example of a closed-loop amplifier with unity feedback factor.

To estimate the open-loop frequency response of the op-amp, let's use the generic model seen in Fig. 21.25. Figure 24.8 shows the location of nodes 1 and 2 on our two-stage op-amp. We've included a load capacitance in this figure. With the help of Table 9.2 we can write

$$R_1 = r_{on} || r_{op} = 111 \text{ k}\Omega$$

$$R_2 = r_{op} || R_{ocasn} \approx r_{op} = 333 \text{ k}\Omega$$

$$g_{m1} = g_{mn} = 150 \text{ }\mu\text{A/V (diff-amp)}$$

$$g_{m2} = g_{mp} = 150 \text{ }\mu\text{A/V (common-source)}$$

$$C_1 = C_{dg4} + C_{gd2} + C_{gs7} = 13.6 \text{ fF}$$

$$C_2 = C_L + C_{gd8} \approx C_L + 1.56 \text{ fF}$$

Let's calculate the open-loop response with a load and compensation capacitance of 100 fF, that is, $C_L = C_c = 100 \text{ fF}$. The pole associated with node 1, from Eq. (21.65), is

$$f_1 \approx \frac{1}{2\pi g_{m2} R_1 R_2 C_c} = 287 \text{ kHz}$$

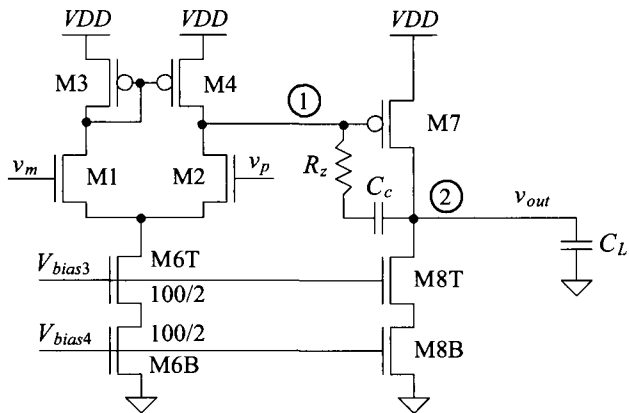


Figure 24.8 Calculating the frequency response of the op-amp.

The location of the pole associated with the output node (node 2) is, from Eq. (21.66),

$$f_2 = \frac{g_{m2}C_c}{2\pi(C_cC_1 + C_1C_2 + C_cC_2)} = 210 \text{ MHz}$$

The location of the zero, f_z in Eq. (21.63), is at 240 MHz (very near the second pole).

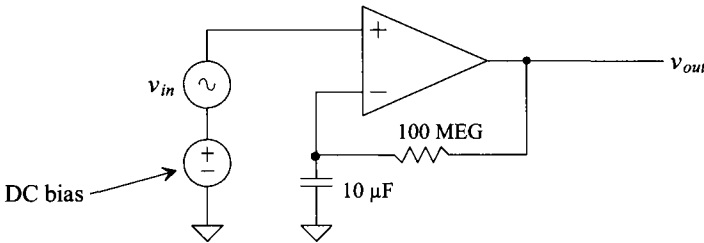


Figure 24.9 Circuit configuration used to simulate open-loop frequency response.

To simulate the open-loop response of the op-amp, we can use the configuration seen in Fig. 24.9. The feedback resistor and capacitor form a time constant so large that for all intents and purposes none of the AC output voltage is fed back to the inverting input. However, the DC bias level is fed back so that the op-amp biases up correctly (all MOSFETs are operating in the saturation region). With a DC bias voltage of 500 mV, Fig. 24.10 shows the open loop responses of the op-amp in Fig. 24.8 with 100 fF compensation capacitance and load capacitance. The simulated values are close to the

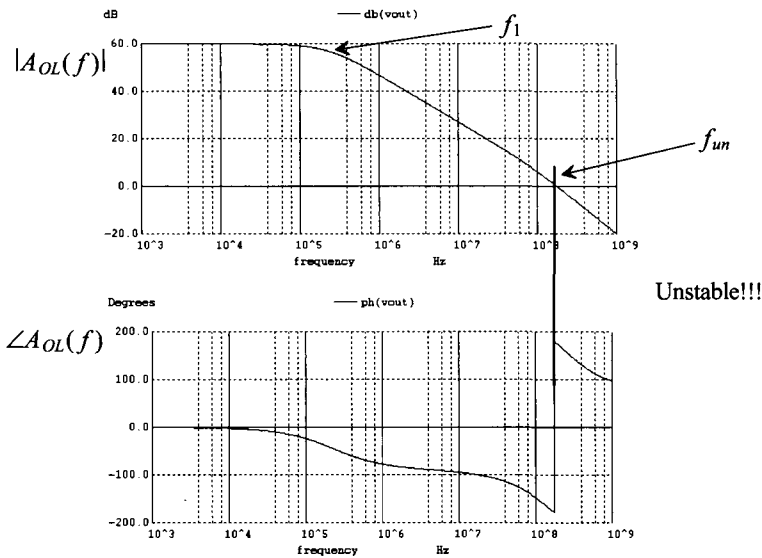


Figure 24.10 The open-loop frequency response of the op-amp in Fig. 24.8 with a 100 fF compensation capacitance and load capacitance.

hand-calculated values given above. However, from the criteria given in Eq. (24.9), the op-amp will clearly be unstable. *We want the open loop gain of the op-amp to be much less than one when the phase shift is 180° .* Looking at Fig. 24.10, we effectively want to shift the lower frequency pole (f_1) downwards in frequency. At the same time, we want to move the higher frequency pole, f_2 , higher in frequency (we want to *split the poles*). Remember this (pole-splitting) was the point of adding the compensation capacitor to an amplifier back in Ch. 21. Let's use Eq. (21.72) to select C_c . Looking at Fig. 24.10, we see that the phase shift is -100° at 10 MHz, so let's set the unity gain frequency to this value (in an attempt to make sure that the open loop gain is well under one when the phase shift is 180°).

$$f_{un} = \frac{g_{m1}}{2\pi C_c} = \frac{150 \mu A/V}{2\pi \cdot C_c} = 10 \text{ MHz} \rightarrow C_c = 2.4 \text{ pF} \quad (24.10)$$

The simulation results are seen Fig. 24.11. The unity-gain frequency, f_{un} , is close to 10 MHz. However, while the lower frequency moved downwards and the higher frequency pole moved upwards, the zero moved down to unity-gain frequency, see Eq. (21.63). This causes the open-loop gain to hover around unity instead of continuing to decrease (this is bad). The stability and step response of the op-amp will be degraded because of the zero. To illustrate this, consider the simulation results seen in Fig. 24.12. Notice that, in a transient simulation, we have to wait some time to let the bias circuit start up. Here, in this simulation, we've waited 500 ns before applying the input signal. Next, notice that our step input signal has a small amplitude change (5 mV). If we apply a larger step, the small-signal analysis used to derive pole-splitting is no longer valid (and we'll see the slew-rate limitations). Clearly the output of the op-amp doesn't behave well with the zero present. To eliminate the zero, we add the zero-nulling resistor, as seen in Fig. 21.33b. Figure 24.11 is regenerated in Fig. 24.13 with the addition of a zero-nulling resistor ($1/g_{mn} = 6.5k$). Figure 24.14 shows the well-behaved step response.

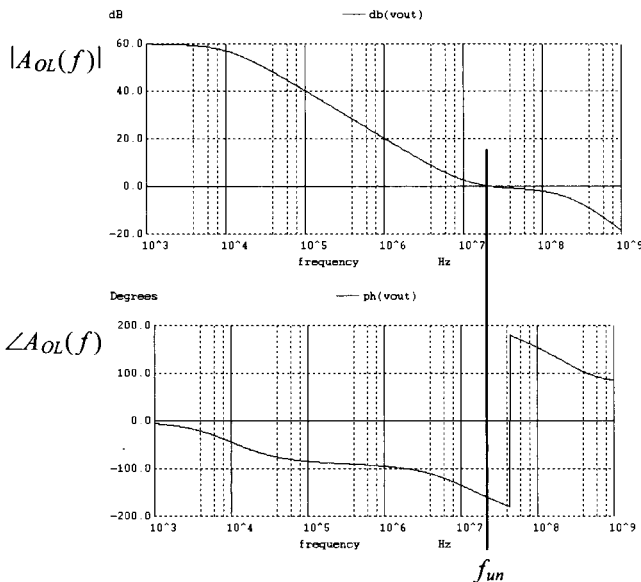


Figure 24.11 Increasing the compensation capacitor's value to 2.4 pF.